

1. Project Overview

Objective

The goal of this project is to build a scalable, enterprise-grade, multi-tenant CRM platform inspired by systems like Salesforce and HubSpot.

The platform will help organizations manage:

- Leads
- Customers
- Sales pipelines
- Contacts
- Accounts
- Workflows & automations
- Permissions & role management
- Team collaboration
- Reporting & analytics

The system is being designed with scalability, extensibility, RBAC security, and workflow automation as core architectural principles.

2. Core CRM Business Model

The CRM follows the standard enterprise CRM lifecycle:

Lead → Account → Contact → Deal

Entity Relationships

Lead

Represents an unqualified prospect.

Examples:

- Website inquiry
- Imported marketing list
- Cold outreach

After qualification, a lead converts into:

- Account
 - Contact
 - Deal
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Account

Represents a company or organization.

Examples:

- Infosys
- TCS
- Google

Stores:

- Company details
- Industry
- Address
- Assigned owner
- Revenue details

Relationship:

- One Account → Multiple Contacts
 - One Account → Multiple Deals
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Contact

Represents a person associated with an Account.

Examples:

- HR Manager
- CTO
- Sales Director

Stores:

- Name
 - Email
 - Phone
 - Job title
 - Linked Account
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Deal / Opportunity

Represents a revenue opportunity.

Examples:

- CRM Implementation – ₹ 2L
- SaaS Subscription Renewal

Stores:

- Deal value
 - Stage
 - Probability
 - Close date
 - Linked contacts
 - Assigned sales owner
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3. Multi-Tenant Architecture

The CRM is designed as a fully multi-tenant platform.

Key Principles

- Each organization has isolated data
 - Tenant context resolved from JWT/session
 - Frontend never sends `tenant_id`
 - Backend enforces tenant isolation
 - Shared infrastructure with logical separation
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4. Role-Based Access Control (RBAC)

The platform includes enterprise-grade RBAC with multiple access layers.

Access Layers

1. Module-Level Access

Controls whether a user can access:

- Leads
 - Deals
 - Contacts
 - Reports
 - Workflows
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2. Action-Level Access

Controls permissions such as:

- Read
- Create
- Update
- Delete

- Export
 - Assign
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3. Record-Level Access

Controls which specific records users can access.

Supported scopes:

- OWN
 - TEAM
 - ALL
 - NONE
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4. Field-Level Security

Controls visibility/editability of fields.

Example:

- Sales reps cannot view deal revenue
 - Managers can access revenue data
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Role Hierarchy

Admin

└─ Manager

└─ Sales Representative

Managers inherit visibility into subordinate data.

5. Team Collaboration Model

The CRM supports team-based collaboration.

Features

- Teams
 - Shared deal visibility
 - Team-based record access
 - Team ownership
 - Cross-functional collaboration
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6. Workflow Automation Engine

One of the core modules is a fully asynchronous workflow engine.

Workflow Architecture

Event → Trigger → Conditions → Actions → Execution

Supported Events

Examples:

- Lead Created
 - Deal Updated
 - Stage Changed
 - Contact Assigned
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Supported Actions

Initial actions include:

- Email notifications
- SMS notifications

- Auto assignment
 - Webhook triggers
 - Internal notifications
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Key Features

- Async processing
 - Retry handling
 - Execution tracking
 - Idempotency protection
 - Delayed/scheduled workflows
 - Extensible action system
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Example Workflow

Scenario

When a deal stage becomes “Qualified”:

1. Assign to Sales Manager
2. Send email notification
3. Trigger webhook to external ERP

7. System Architecture

Backend Stack

Planned Technologies

Layer	Technology
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Backend Framework	Spring Boot
Security	Spring Security + JWT
Database	PostgreSQL
Cache	Redis
Queue System	RabbitMQ / Kafka
ORM	Hibernate / JPA
API Style	REST APIs

Frontend Stack

Layer	Technology
Framework	Next.js
UI	React

Styling Tailwind CSS

State TanStack
Management Query

Authentication JWT

9. Security Considerations

The system is designed with enterprise security practices.

Security Features

- JWT authentication
- Tenant isolation
- RBAC enforcement
- Record-level access filtering
- Field masking
- Audit logging
- Secure webhook validation
- Rate limiting

10. Scalability & Performance

The architecture is designed for future scale.

Scalability Features

- Async workflow execution
- Redis caching
- Queue-based event processing
- Stateless APIs

- Modular services
- Database indexing strategy
- Extensible action processors

11. Development Goals

Current Goals

- Build scalable CRM foundation
 - Implement enterprise RBAC
 - Develop workflow automation engine
 - Create reusable API architecture
 - Design extensible module structure
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12. Project Vision

The long-term vision is to create a modular enterprise CRM platform capable of supporting:

- SMB organizations
- Mid-sized enterprises
- Multi-team sales operations
- Workflow-driven automation

13. Review & Feedback Requested

Feedback requested on:

- Business workflows
- Tenants
- Required modules
- Approval process

- Team structures
- Workflow automation requirements
- Reporting expectations
- Integration requirements
- Feature prioritization